

EC412 Satellite Communication Systems (3 Credits)

Prerequisites: EC254 Transmission Lines and Waveguides, EC302 Digital Communication Systems.

Course Learning Outcomes: Students will be able to design satellite link budgets, calculate carrier-to-noise ratio (C/N), and analyze uplink and downlink path loss across Ku-band (12-18 GHz) and Ka-band (26-40 GHz) transponders. Module 3 covers Geostationary Earth Orbit (GEO) and Low Earth Orbit (LEO) satellite constellation dynamics.

EC413 Radar and Navigation Systems (3 Credits)

Prerequisites: EC301 Electromagnetic Theory, EC352 Digital Signal Processing.

Course Learning Outcomes: Analyze pulsed Doppler radar operation, maximum unambiguous range equations, and synthetic aperture radar (SAR) imaging algorithms for aerospace navigation.